Adaptive Defense Evaluation - Deliverable 0

Christian Farrell

1. Assigned project and research question

Project Title: Adaptive Defense Evaluation with Security-Utility Tradeoffs

This project aims to identify whether language-model guards/defenses still hold up even when an attacker observes the defense and adapts their attack. Many of these model guards are tested by static attack methods, however, do these defenses fall apart when the attacker can adapt?

This problem matters for language-model security because a defense which performs well against fixed attacks might not hold up against a real attacker who can change their strategy. At the same time a model should not refuse asinine requests in favor of blocking would-be attacks.

Research Question: Under a fixed attack budget, how much does the security of language model erode when an attacker can observe a defense and adapt their attacks, and how does this anticipation affect benign requests and false refusals?

2. Smallest useful study

The smallest possible useful study would include the following.

Input: A set of harmful prompts and their corresponding attack method, and a set of benign and their corresponding request category.

Output: The target language model's response to each attack, as well as if it was successful or not. Also recording whether or not benign prompts had been unnecessarily refused and why.

Baseline: Each selected open model without an added defense, evaluated using the same fixed attack and benign-task sets.

Main Comparison: A defensive model evaluated against both static attacks and an adaptive attacker that can use the defense's previous responses to modify subsequent attacks.

Main Measure: Attack Success Rate (ASR), compares static and adaptive attack successfulness.

3. Dependency map

Dependency area	What you need	Why it is needed / what depends on it	Current status	Next action
Data or evaluation set	A fixed subset of HarmBench/StrongREJECT/etc containing both	Both static and adaptive evaluations should use the	Needs confirmation	Need to confirm with prof to see if these are the best

	harmful and benign prompts	same prompts so we can definitively attribute differences to the defenses/attacker conditions.		benchmarks or if I should use something else.
Model, tokenizer, system, or tool	Two approved open models + their tokenizers + 2-3 defenses	We need to compare defenses under both static and adaptive attacks.	Needs confirmation	Identify the exact two open models and the available defenses, then verify that they can run locally.
NLP or research method	Static language-model attacks, adaptive language-model attacks	Our central research question depends on comparing these attacks.	Partly familiar	Read my starting source in depth and do additional research.
Compute, software or access	GPUs capable of running selected models + inferences libraries	Project requires repeated model calls for static and adaptive attacks, insufficient compute could prevent progress.	Needs confirmation	Need to identify models first, then evaluate options.
Evaluation, human review, or safety control	ASR, FRR, Query Cost, Human Review	ASR alone could make a defense that simply refuses everything appear successful. Need to evaluate both ASR and FRR, alongside other important metrics like cost and speed.	Partly available	Define the exact metrics and labeling rules, then determine which harmful-success and false-refusal examples will receive human review.

4. Initial evidence and evaluation plan

The **primary measure** we will use for evaluating success will be Attack Success Rate (ASR), which measures the percentage of attacks that successfully cause the model to produce the targeted harmful behavior. A meaningful result would show that the security of a model decreased when the attacker was able to adapt to defenses.

An additional measure I will use is False-Refusal Rate (FRR) which records the percentage of benign requests that were falsely refused as a defense attempt. This prevents a defense from appearing more secure simply because it refuses a large number of harmless requests. I will also record query count and latency to measure how much more adaptive attacks require substantially more resources compared to static attacks.

In order to enable someone to check the result later the following information should be saved:

- Exact prompts and attack conditions
- Model and defense configurations
- Attack budget
- Adaptive queries and model responses
- Success/failure labels
- False-refusal labels
- Latency
- Query counts

5. Main risk, backup scope, and immediate next step

The **main risk** of this current project plan is not having enough compute to host the chosen models.

A **smaller scope** would be to reduce the study to one open model, one attack family, and two defenses. This would still answer the main research question, just with fewer datapoints.

My **immediate next step** is to verify which open models, defenses and attack datasets I should use.

6. Starting source and AI Use Statement

Starting Source: Jailbreaking Leading Safety-Aligned LLMs with Simple Adaptive Attacks
Andriushchenko, M., Croce, F., & Flammarion, N. *Jailbreaking Leading Safety-Aligned LLMs with Simple Adaptive Attacks*. ICLR 2025.

This paper showcases exactly what I am working towards with this project. It shows that language models can be jailbroken when the attacker adapts their attack strategy. This supports my project's focus on comparing static attacks with adaptive attacks and understanding whether or not defenses remain effective when attacks become adaptive.

AI Use Statement:

Tool and model used. N/A
Approximate date or dates of use. N/A
Purpose of use. N/A
Sections affected. N/A
How you checked the resulting text, claims, citations, or plan. N/A